CPE 314 — Software Design

University of Cebu

Computer Engineering Department

Plate N Platter

Plan smarter. Eat better. Cook easier

Project Design Document

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Table of Contents

Revision History....3
1. Introduction & Project Overview....4
2. Preliminary ER Sketch & Team Roles....5
3. System Architecture & Object-Oriented Design....7
4. Data Model & Database Design....7
5. API Integration Design....7
6. Concurrency Design....7
7. Data Mining & Visualization Design....8
8. Testing Summary....8
9. Individual Contributions....8
Appendix....9

Revision History

Update this table each week as your team adds or revises sections of this document.

Version	Date	Author	Changes
0.1	October 22, 2026	GROUP PNP	Initial draft — Week 8 sections
0.2			Week 9 sections added
0.3			Week 10 sections added
1.0			Final compilation and polish — Week 11

Formatting Guidelines

Follow these formatting rules consistently throughout the document so the final compiled version reads as one polished deliverable, not four stitched-together drafts.

Element	Rule
Body font	Calibri, 11pt (the default font of this template — do not mix fonts)
Headings	Use Word's built-in Heading 1/2/3 styles (already applied to every section title below) — this is required for the Table of Contents field to work
Margins	1 inch on all sides
Line spacing	1.15–1.5 line spacing for body paragraphs; single spacing inside tables
Table captions	Label every table "Table N: short description" directly ABOVE the table, numbered sequentially from Table 1 through the end of the document
Figure/chart captions	Label every chart/screenshot "Figure N: short description" directly BELOW the figure, numbered sequentially from Figure 1
Code snippets	Use a monospace font (Consolas or Courier New), 9– 10pt, and keep each snippet under ~20 lines — link to the full file in your repository instead of pasting it all

Page numbers	Already automatic — every page footer includes a live page number field
Alignment	Body text left-aligned/justified; headings left-aligned; do not center body paragraphs

1. Introduction & Project Overview

Plate N Platter is a personal meal-planning desktop application for individual users such as students and busy households, built for Option H (Recipe & Meal Planner) of CPE 314. Users register, log in, and plan breakfast, lunch, dinner, and snacks on a calendar where every meal is color-coded by Healthy Level (Healthy, Moderate, or Indulgent) and labeled Quick, Easy, or Hard to prepare. Recipes can be imported from TheMealDB (including Filipino dishes) or created by the user, can be built from other recipes, and feed an automatic shopping list, a cook-next queue, an undo function, and a daily streak that rewards eating at least one healthy meal a day. The app solves the everyday problem of deciding what to cook, knowing what to buy, and staying consistent with healthier eating, all in one simple tool.

Table 2 : Project Summary

ITEM	DETAILS
Theme	Option H — Recipe & Meal Planner
Target users	Individual users (students, busy households); each user has a private account
Platform	Java 17+ desktop application with a JavaFX graphical interface
Data storage	SQLite database accessed through JDBC using PreparedStatement
Public API	TheMealDB (https://www.themealdb.com/api.php), shared test key “1”
Key features	Register/login, calendar with Healthy Level colors, today’s meals with photos, recipe browsing and creation, nested recipes, cook queue, undo, shopping list, streak and badges, weekly statistics with a bar chart

2. Preliminary ER Sketch & Team Roles

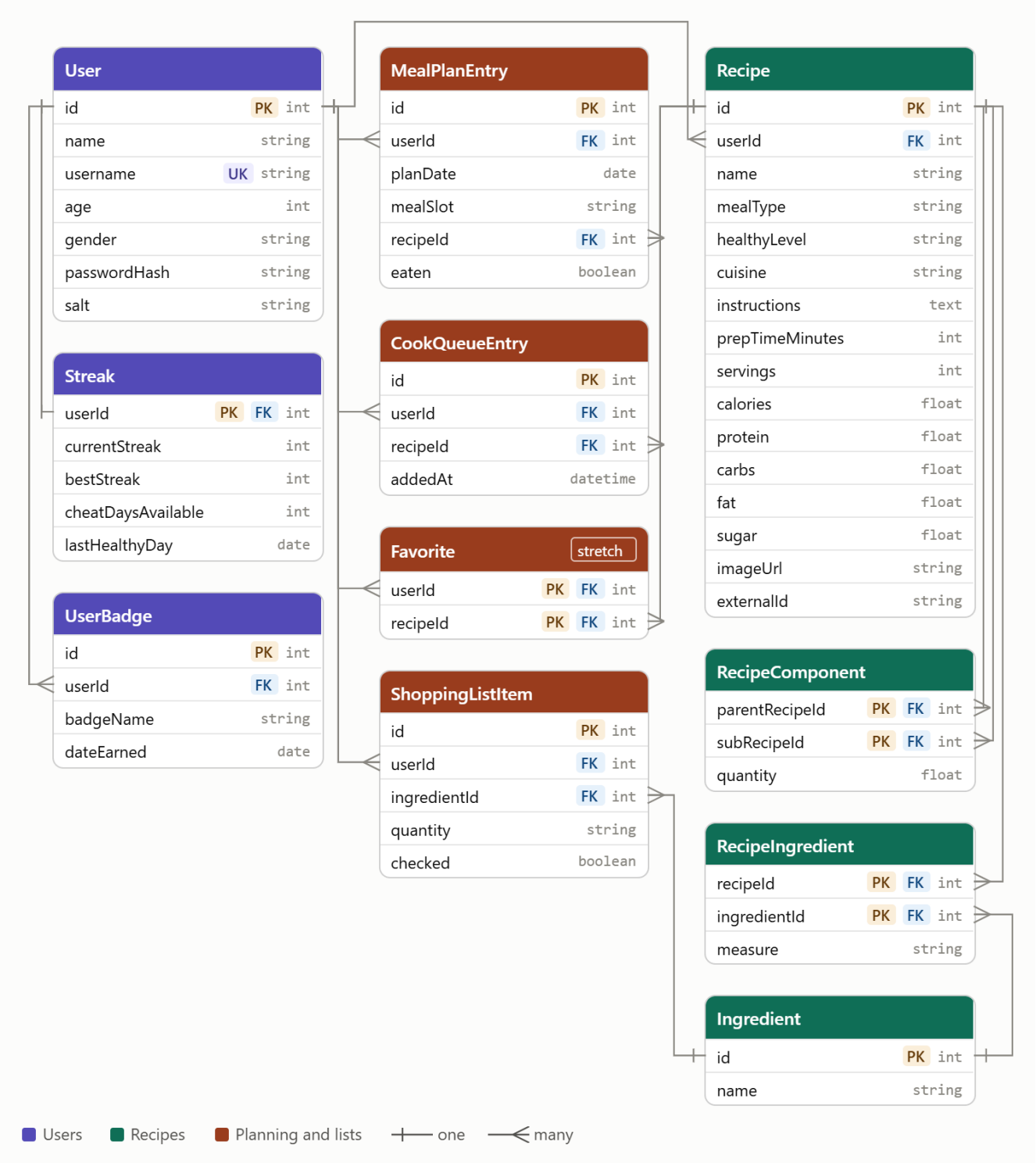


Figure 1: Preliminary ER sketch (entities and key attributes)

Table 3: Relationship in the preliminary sketch

From	To	Cardinality	Meaning
Users	Recipes	1 : M	A user creates many recipes
Users	MealPlanEntries	1 : M	A user plans many meals on the calendar
Recipes	MealPlanEntries	1 : M	A recipe can be planned many times
Recipes	Ingredients	M : N	Linked through RecipeIngredient
Recipes	Recipes	M : N	A recipe uses sub-recipes, linked through RecipeComponent
Users	ShoppingListItems	1 : M	Each user has a shopping list
Users	CookQueueEntries	1 : M	Each user has a cook queue
Users	Streaks	1 : 1	One streak record per user
Users	UserBadges	1 : M	A user earns many badges
Users	Recipes (Favorite)	M : N	A user saves favorite recipes (stretch)

Table 4:Team members and primary roles

Member	Primary Role
Kimberly Luistro	Team Leader
Nesley Cate Librea	UI
Venice Nadela	QA
Yvonne Mae Tapan	UI
Kirk Quisagan	Developer
Ken Tamidles	DB admin

3. System Architecture & Object-Oriented Design

Present your class hierarchy (a diagram or a clear textual description) showing the abstract class/interface and its concrete subclasses, and justify how each class follows the Single Responsibility Principle. Explain the design rationale in a short paragraph.

4. Data Model & Database Design

Include the finalized ER diagram (3+ entities, at least one relationship with correct cardinality) and the actual SQLite schema (CREATE TABLE statements or an equivalent schema table). This section supersedes the preliminary sketch from Section 2.

5. API Integration Design

Name the public API your app consumes, describe the endpoint(s) called, the shape of the JSON response, and how that data is parsed and used meaningfully in the app.

6. Concurrency Design

Identify the long-running operation(s) that run on a background thread, explain the threading approach used, and describe how the UI stays responsive during that operation.

7. Data Mining & Visualization Design

Describe the aggregation or frequency-analysis step performed on your app's data and the chart(s) used to visualize the result, including why that chart type was chosen.

8. Testing Summary

Summarize your JUnit test suite — how many tests exist, what they cover, and at least one meaningful edge case per tested class. A short table or list of key test cases is helpful here.

9. Individual Contributions

List what each team member specifically built or owned, referencing their git commit history, so the instructor can cross-check contribution during the Individual Q&A.

Appendix

(Add any additional references, links to full API documentation, licensing notes, or supplementary material here.)